

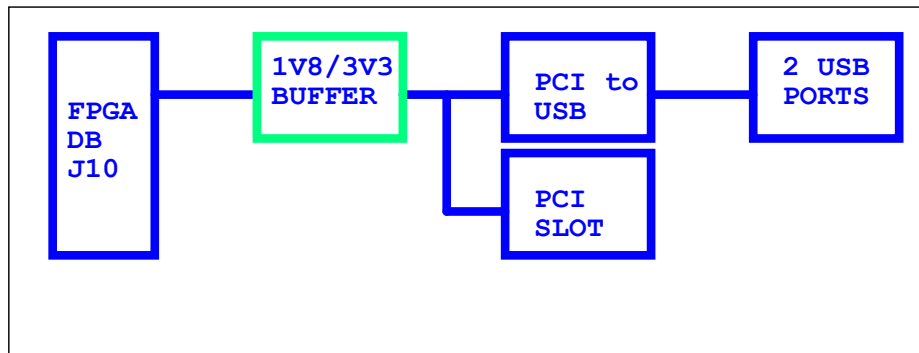
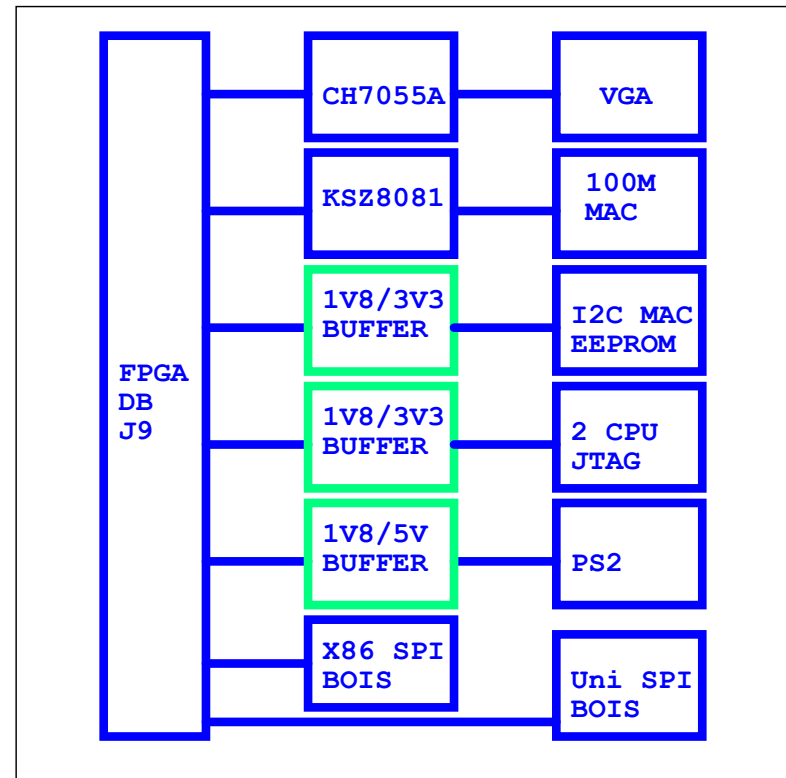
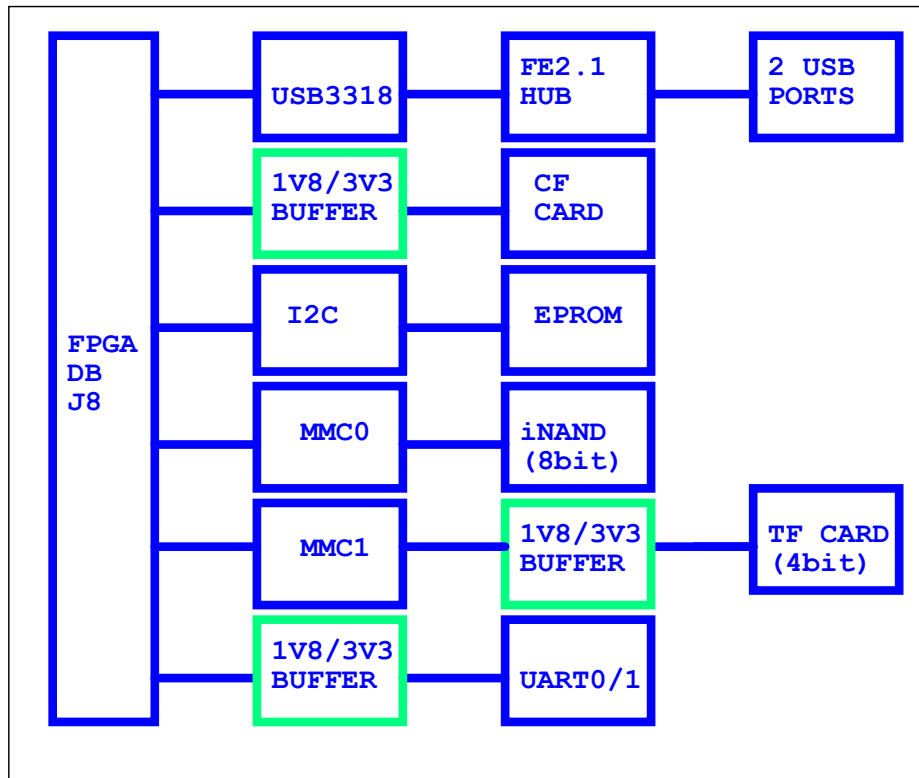
# PKUnity Microsystems Technology CO.,LTD

## Three V7-FPGA CARD-V1.2 Schematic

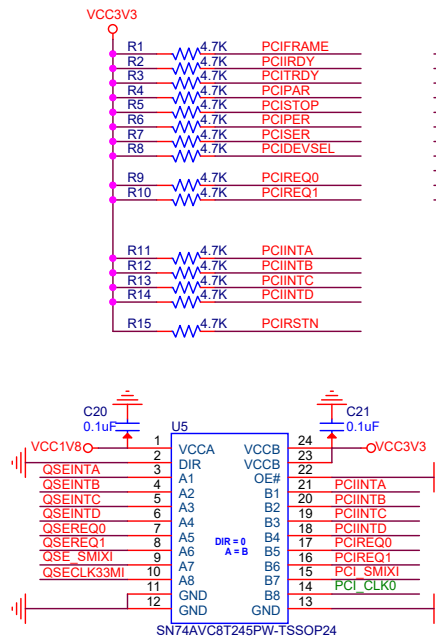
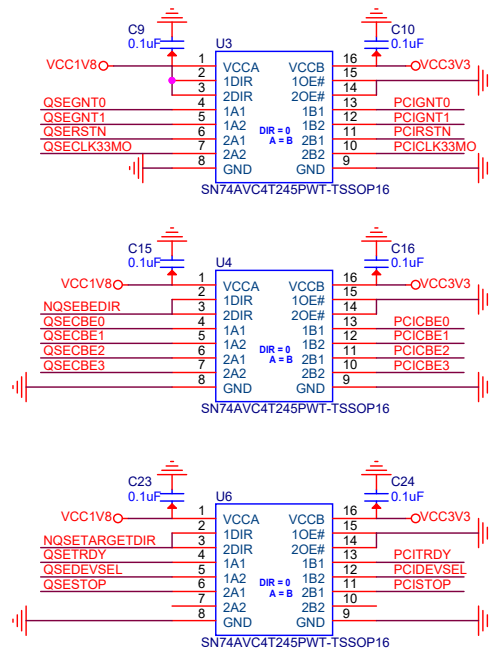
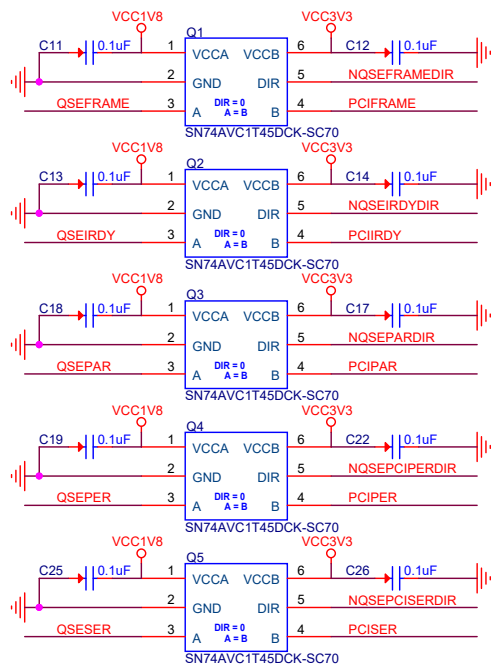
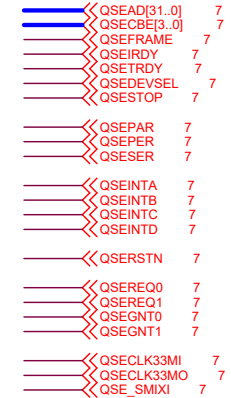
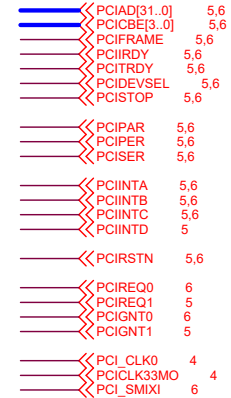
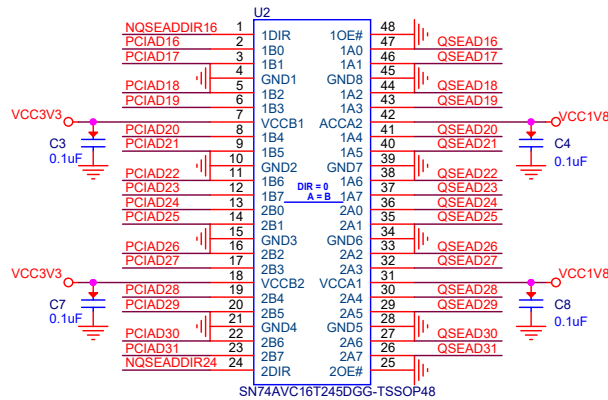
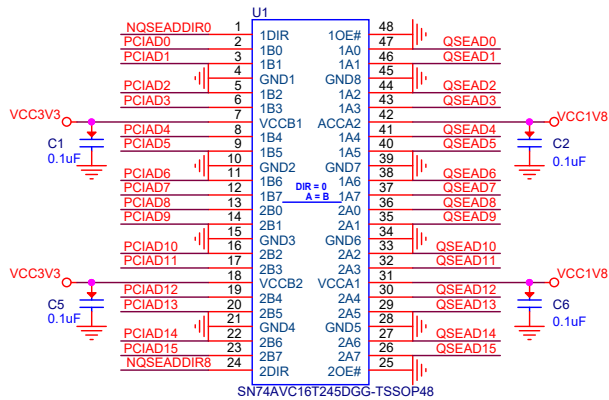
### Page Index

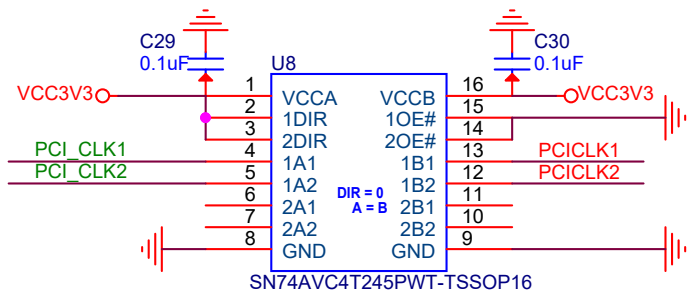
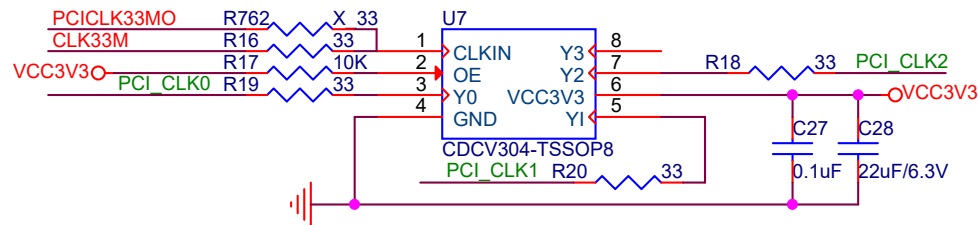
|                     |
|---------------------|
| 01-Cover Page       |
| 02-Block Diagram    |
| 03-B1_PCI BUFFER    |
| 04-B1_PCI CLOCK     |
| 05-B1_PCI-SLOT      |
| 06-B1_PCItOUSB      |
| 07-B1_S2C_DB1       |
| 08-B1_POWER         |
| 09-B2_SPI EPROM     |
| 10-B2_CPU ICE       |
| 11-B2_PS2           |
| 12-B2_VGA           |
| 13-B2_MII LAN(100M) |
| 14-B2_S2C_DB2       |
| 15-B3_IDE-BUFFER    |
| 16-B3_CF CARD       |
| 17-B3_UART I2C      |
| 18-B3_iNAND TF-CARD |
| 19-B3_USB PHY       |
| 20-B3_USB HUB       |
| 21-B3_S2C_DB3       |

|                                                                                                                                                                                                                                                                                                            |                          |         |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|---------|--|
|                                                                                                                                                                                                                       |                          |         |  |
| Page Name COVER PAGE                                                                                                                                                                                                                                                                                       |                          |         |  |
| Size C                                                                                                                                                                                                                                                                                                     | Project Name V7FPGA CARD | Rev 1.2 |  |
| Date: Friday, May 26, 2023                                                                                                                                                                                                                                                                                 | Sheet 1 of 21            |         |  |
| <small>PROPERTY NOTE: this document contains information confidential and property to TOPSTAR and shall not be reproduced or transferred to other documents or disclosed to others or used for any purpose other than that for which it was obtained without the expressed written consent of MPRC</small> |                          |         |  |

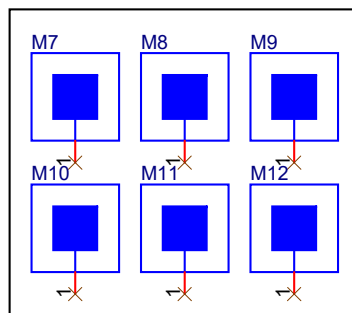
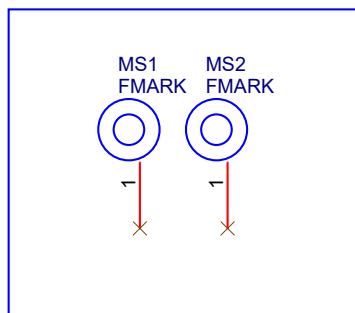
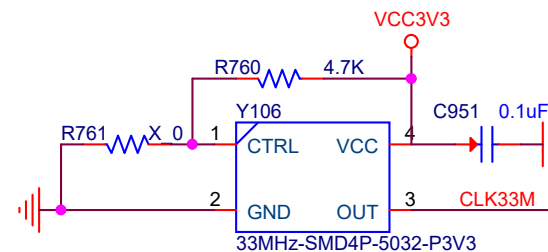


|                                                                                                                                                                                                                                                                                             |              |               |         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---------------|---------|
| Page Name                                                                                                                                                                                                                                                                                   |              | Block Diagram |         |
| Size B                                                                                                                                                                                                                                                                                      | Project Name | V7FPGA CARD   | Rev 1.2 |
| Date: Friday, May 26, 2023                                                                                                                                                                                                                                                                  |              | Sheet         | 2 of 21 |
| PROPERTY NOTE: this document contains information confidential and property to TOPSTAR and shall not be reproduced or transferred to other documents or disclosed to others or used for any purpose other than that for which it was obtained without the expressed written consent of MPRC |              |               |         |



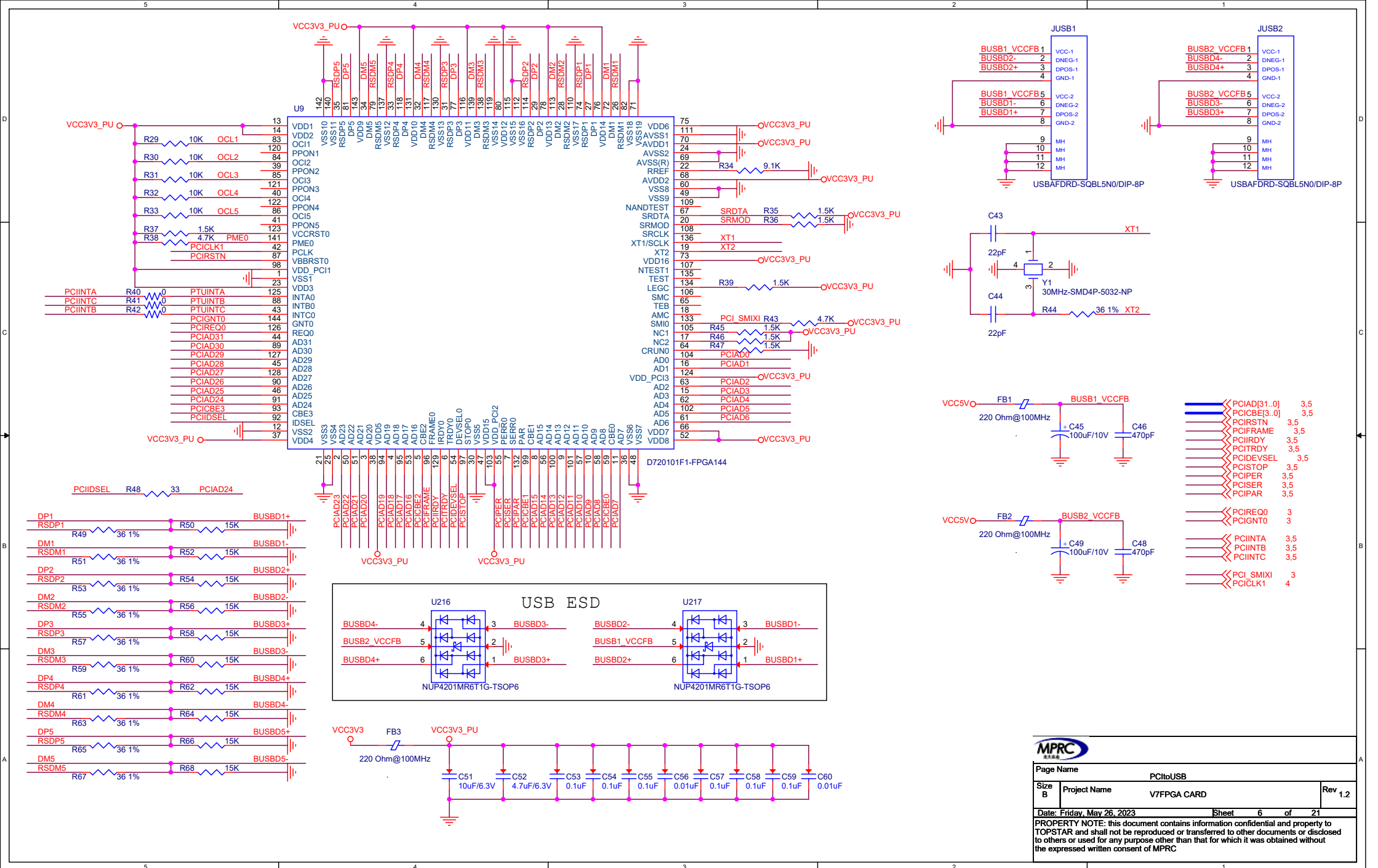


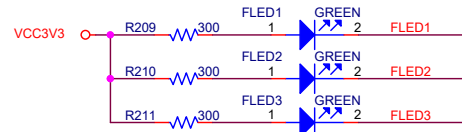
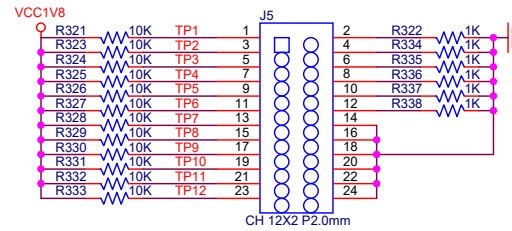
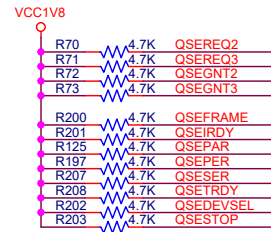
PCI CLOCK 统一 buffer 延时



|                                                                                                                                                                                                                                                                                             |              |         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---------|
|                                                                                                                                                                                                        |              |         |
| Page Name                                                                                                                                                                                                                                                                                   |              |         |
| PCI CLOCK                                                                                                                                                                                                                                                                                   |              |         |
| Size A                                                                                                                                                                                                                                                                                      | Project Name | Rev 1.2 |
| V7FPGA CARD                                                                                                                                                                                                                                                                                 |              |         |
| Date: Friday, May 26, 2023                                                                                                                                                                                                                                                                  |              |         |
| Sheet 4 of 21                                                                                                                                                                                                                                                                               |              |         |
| PROPERTY NOTE: this document contains information confidential and property to TOPSTAR and shall not be reproduced or transferred to other documents or disclosed to others or used for any purpose other than that for which it was obtained without the expressed written consent of MPRC |              |         |

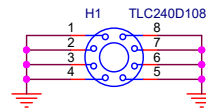


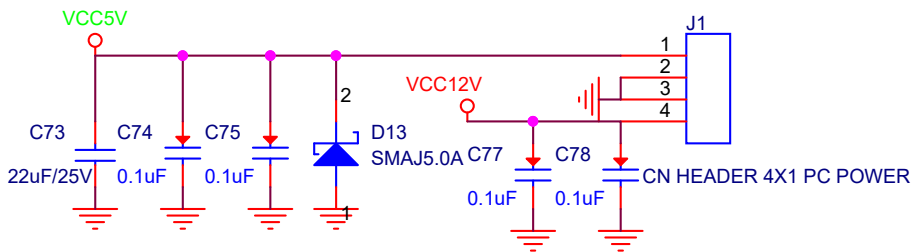
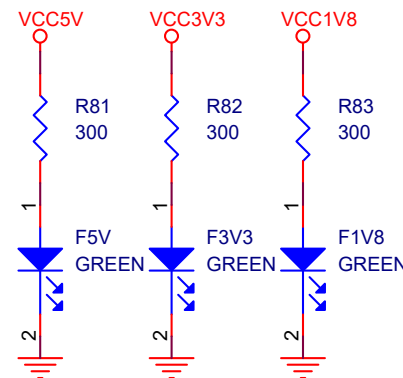
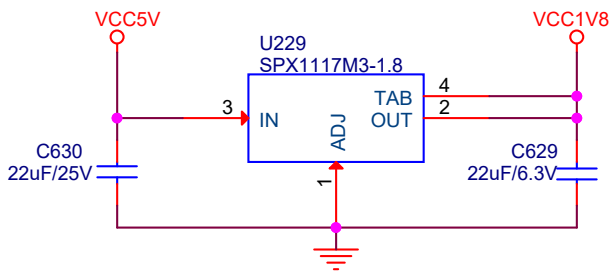
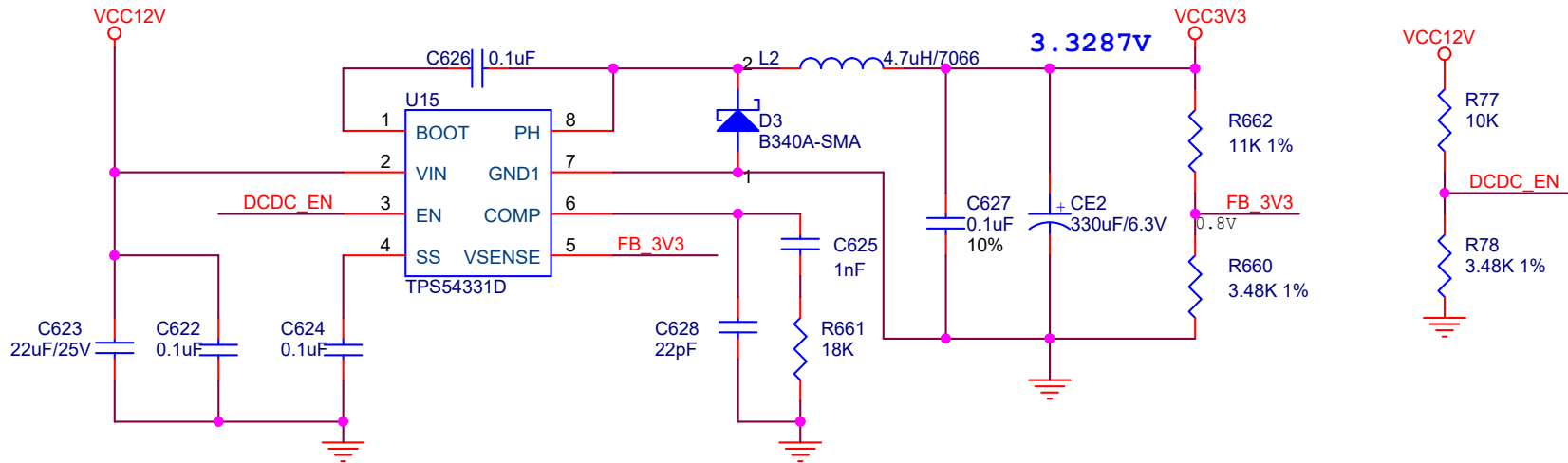




YK CLOCK: P/N = 4/3; 63/61; 68/66

S2C CLOCK: P/N = 36/34; 63/61; 64/62;100/98  
S2C SLOT : J3/J4/J5





Page Name

POWER

Size  
A

Project Name

V7FPGA CARD

Rev 1.2

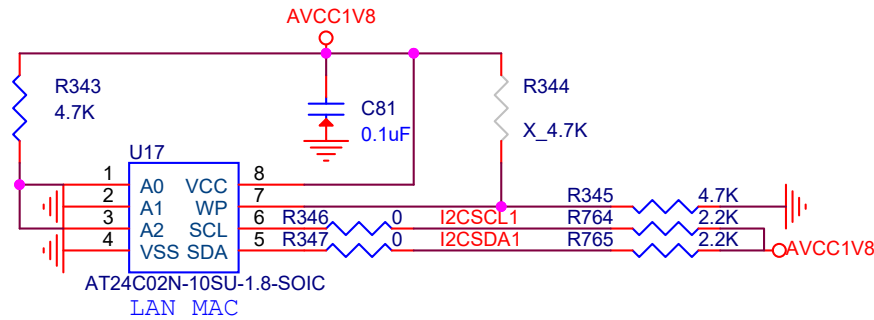
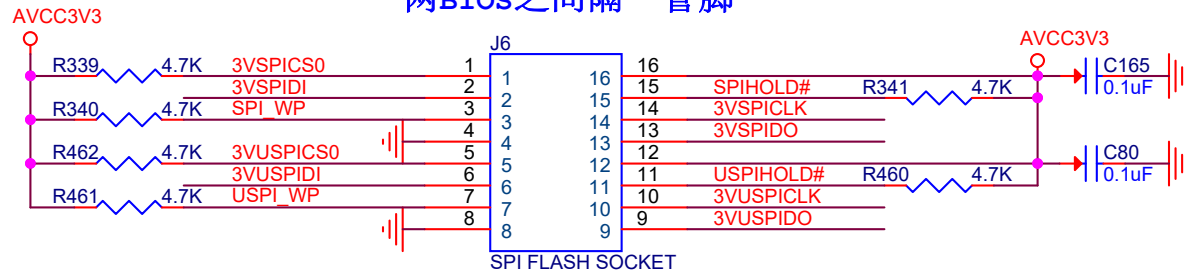
Date: Friday, May 26, 2023

Sheet 8 of 21

PROPERTY NOTE: this document contains information confidential and property to TOPSTAR and shall not be reproduced or transferred to other documents or disclosed to others or used for any purpose other than that for which it was obtained without the expressed written consent of MPRC



双BOIS SPI FLASH 插座  
注意检查PCB封装:  
两BIOS之间隔一管脚



|           |    |
|-----------|----|
| 3VSPIDO   | 10 |
| 3VSPIDI   | 10 |
| 3VSPICLK  | 10 |
| 3VSPICS0  | 10 |
| 3VUSPIDO  | 10 |
| 3VUSPIDI  | 10 |
| 3VUSPICLK | 10 |
| 3VUSPICS0 | 10 |
| I2CSDA1   | 14 |
| I2CSCL1   | 14 |



Page Name

SPI MAC ROM

Size  
A

Project Name

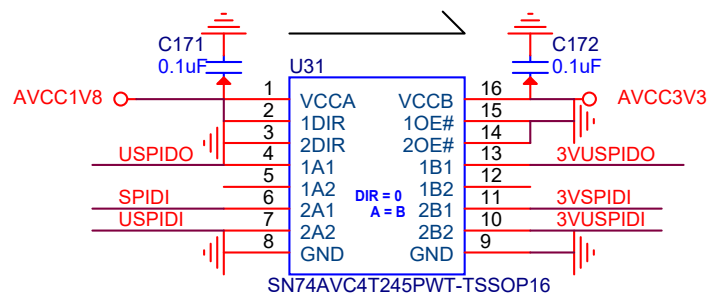
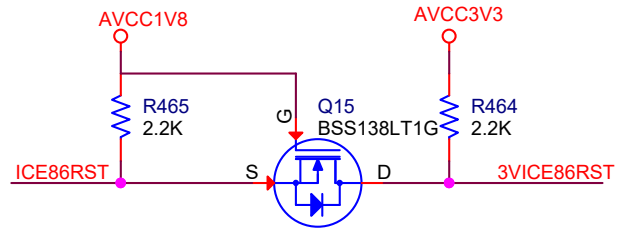
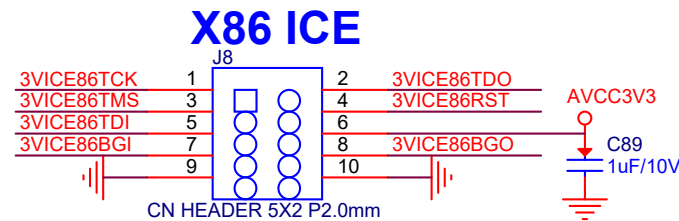
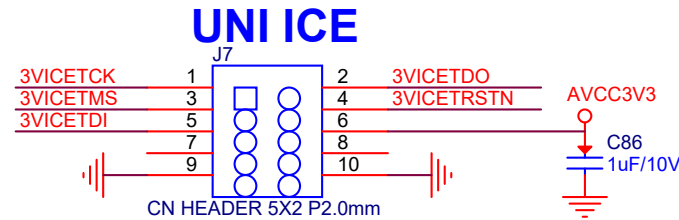
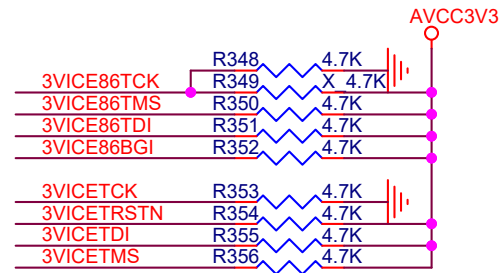
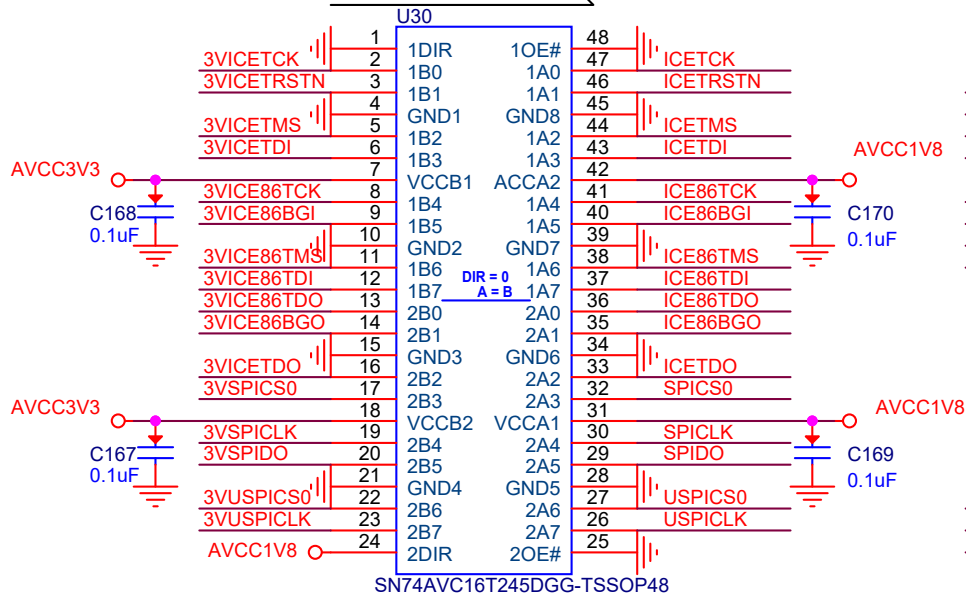
V7FPGA CARD

Rev 1.2

Date: Friday, May 26, 2023

Sheet 9 of 21

PROPERTY NOTE: this document contains information confidential and property to TOPSTAR and shall not be reproduced or transferred to other documents or disclosed to others or used for any purpose other than that for which it was obtained without the expressed written consent of MPRC



Page Name

CPU ICE

Size

Project Name

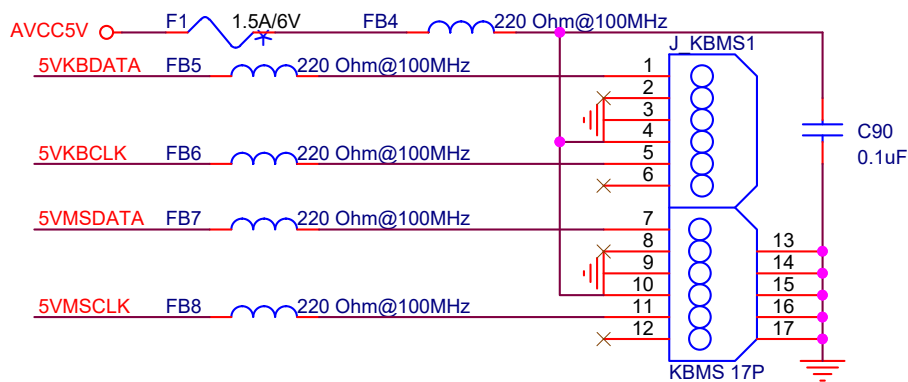
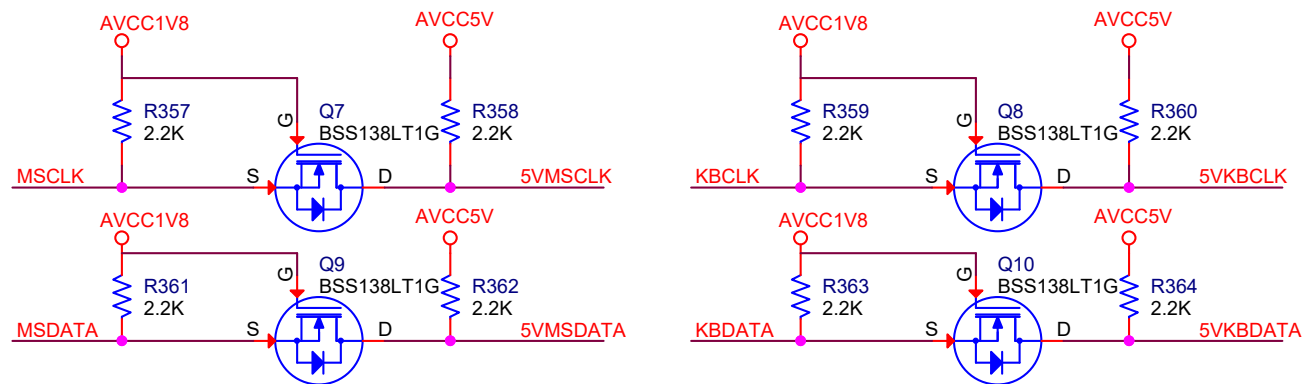
V7FPGA CARD

Rev 1.2

Date: Friday, May 26, 2023

Sheet 10 of 21

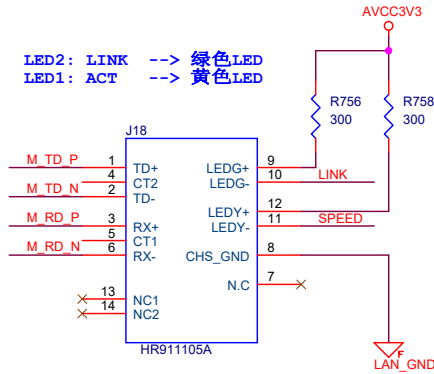
PROPERTY NOTE: this document contains information confidential and property to TOPSTAR and shall not be reproduced or transferred to other documents or disclosed to others or used for any purpose other than that for which it was obtained without the expressed written consent of MPRC



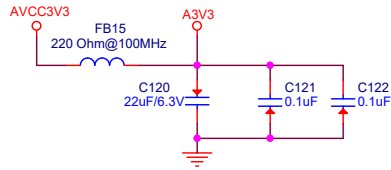
|                                                                                                                                                                                                                                                                                             |                          |         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|---------|
|                                                                                                                                                                                                        |                          |         |
| Page Name PS2                                                                                                                                                                                                                                                                               |                          |         |
| Size A                                                                                                                                                                                                                                                                                      | Project Name V7FPGA CARD | Rev 1.2 |
| Date: Friday, May 26, 2023 Sheet 11 of 21                                                                                                                                                                                                                                                   |                          |         |
| PROPERTY NOTE: this document contains information confidential and property to TOPSTAR and shall not be reproduced or transferred to other documents or disclosed to others or used for any purpose other than that for which it was obtained without the expressed written consent of MPRC |                          |         |



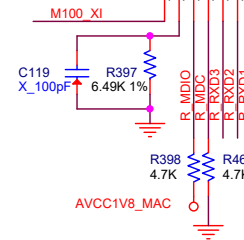
LED2: LINK --> 绿色LED  
LED1: ACT --> 黄色LED



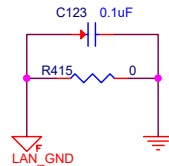
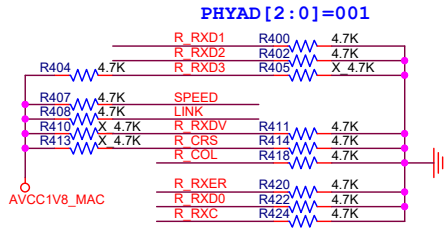
差分走线为100欧阻抗



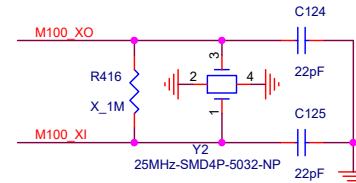
R397, C119  
尽量靠近  
芯片摆放



MACTXD[3..0] 14  
MACRXD[3..0] 14  
MACTXEN 14  
MACRXDV 14  
MACRXER 14  
MACCRS 14  
MACCOL 14  
MACMDIO 14  
MACMDC 14  
MACTXCLKI 14  
MACRXCLKI 14  
ASYSRSTN 14

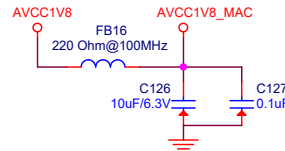


LAN\_GND必须接PCB GND

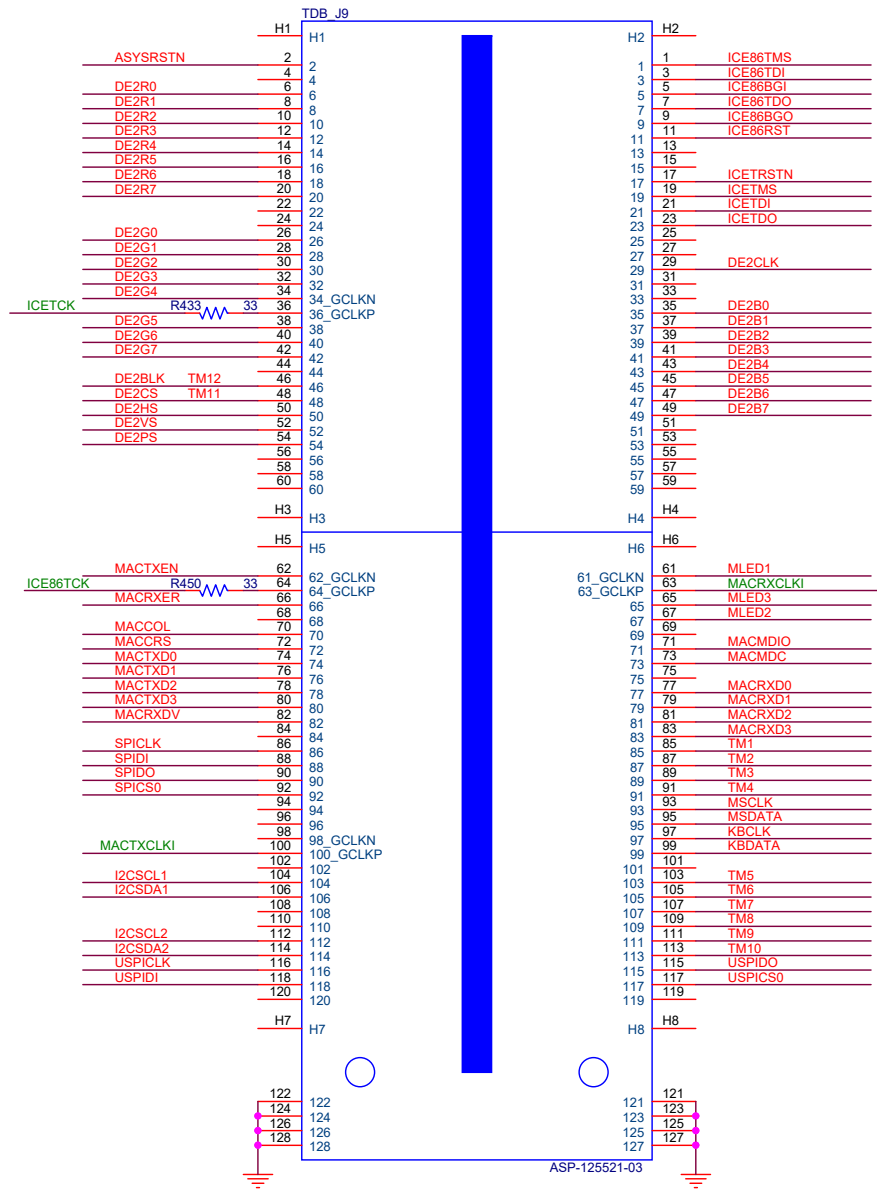


R\_RXD0 MACRXD0  
R\_RXD1 MACRXD1  
R\_RXD2 MACRXD2  
R\_RXD3 MACRXD3  
R\_RXDV MACRXDV  
R\_RXER MACRXER  
R\_CRS MACCRS  
R\_COL MACCOL  
R\_RXC R421 33 MACRXCLKI  
R\_TXC R423 33 MACTXCLKI  
R\_MDC R425 33 MACMDC  
R\_MDIO R426 33 MACMDIO  
T\_TXD0 MACTXD0  
T\_TXD1 MACTXD1  
T\_TXD2 MACTXD2  
T\_TXD3 MACTXD3  
T\_TXEN MACTXEN

| SET              | Function                  |
|------------------|---------------------------|
| [RXDV, CRS, COL] | 000 MII                   |
|                  | 110 MII Back to Back      |
| R_RXER           | 0 Disable ISOLATE         |
| SPEED            | 1 100M                    |
| LINK             | 1 Enable Auto-Negotiation |
| R_RXD0           | 0 Full Duplex             |
| R_RXC            | 0 Broadcast PHY address   |

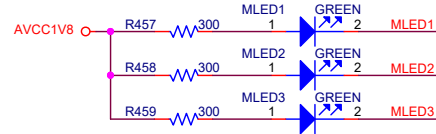


|                                                                                                                                                                                                                                                                                             |                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| Page Name                                                                                                                                                                                                                                                                                   | MII LAN(100M)            |
| Size B                                                                                                                                                                                                                                                                                      | Project Name V7FPGA CARD |
| Date: Friday, May 26, 2023                                                                                                                                                                                                                                                                  | Sheet 13 of 21           |
| PROPERTY NOTE: this document contains information confidential and property to TOPSTAR and shall not be reproduced or transferred to other documents or disclosed to others or used for any purpose other than that for which it was obtained without the expressed written consent of MPRC |                          |

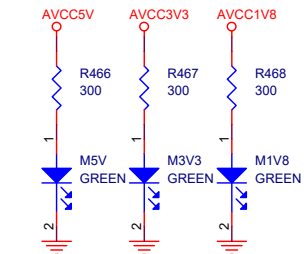
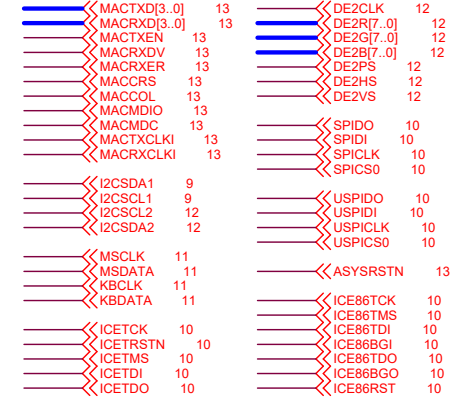
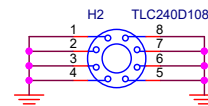
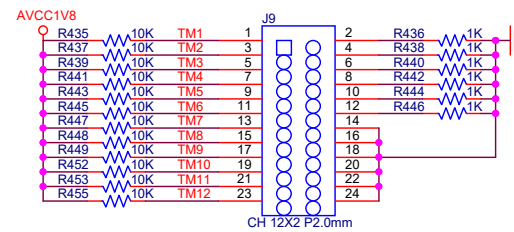
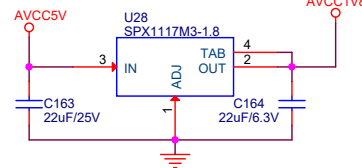
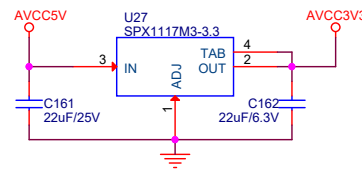
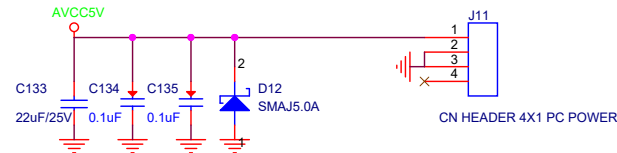


YK CLOCK: P/N = 4/3; 63/61; 68/66

S2C CLOCK: P/N = 36/34; 63/61; 64/62; 100/98  
S2C SLOT : J3/J4/J5

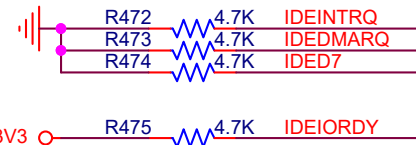
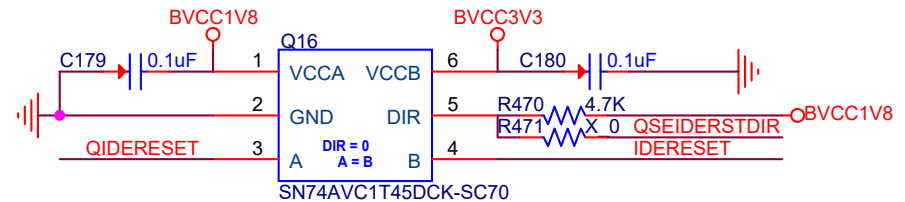
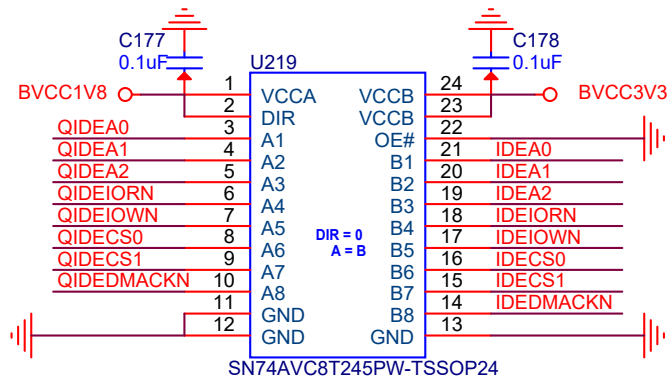
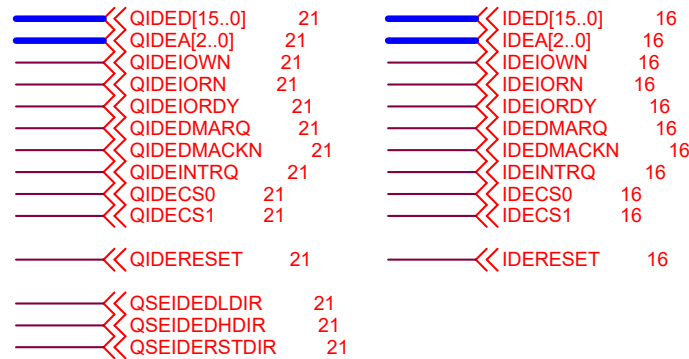
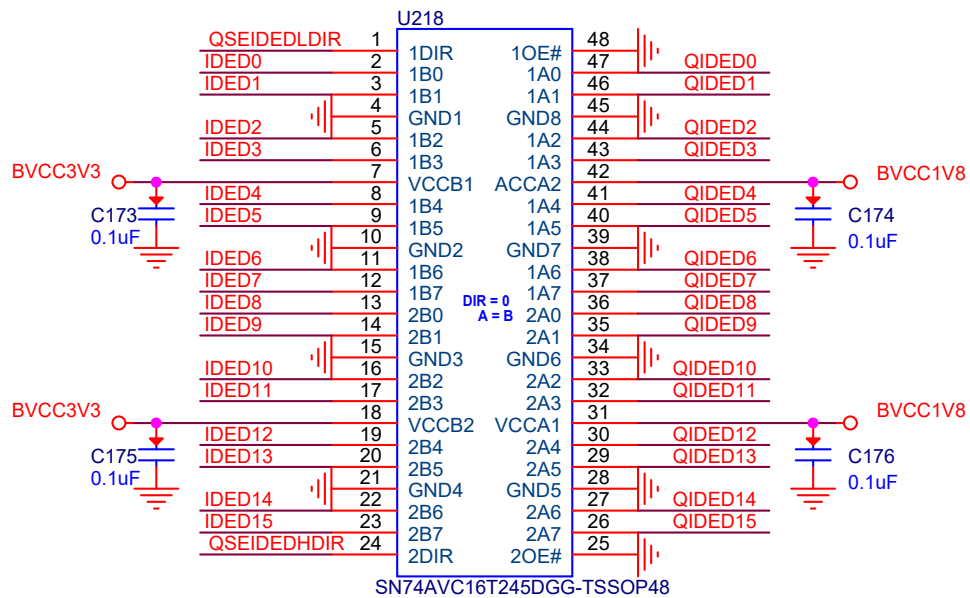


英菲FPGA板不需要外接12V  
S2C和亚科板必须外接12V



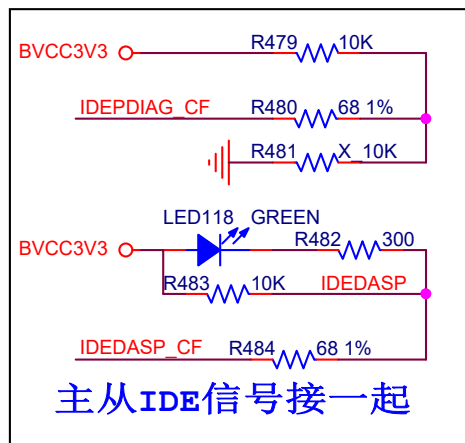
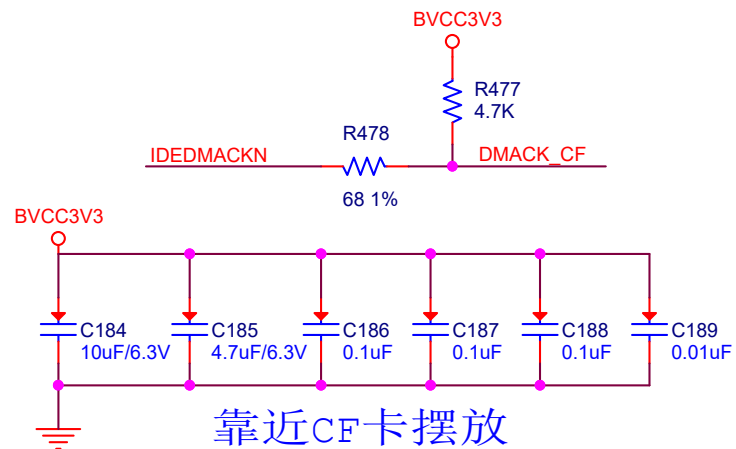
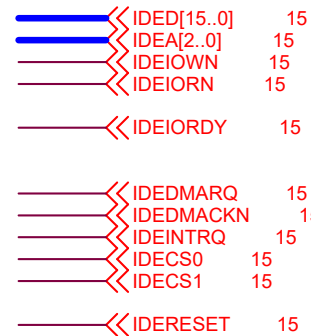
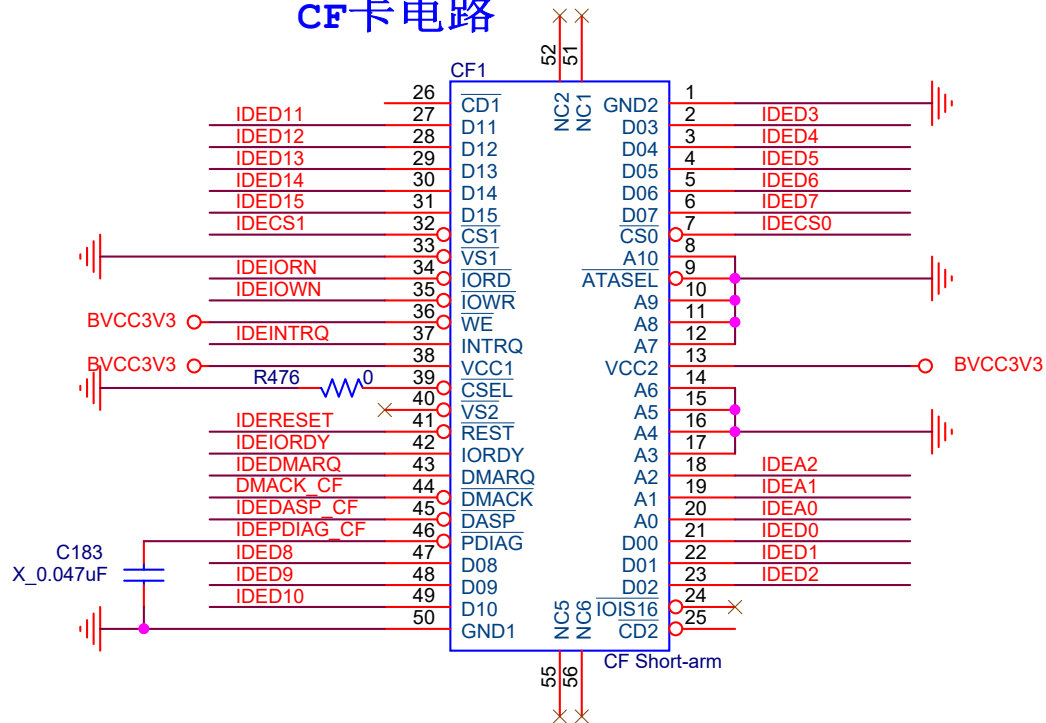
|                            |              |          |
|----------------------------|--------------|----------|
| Page Name                  | S2C-DB2      |          |
| Size B                     | Project Name | Rev 1.2  |
|                            | V7FPGA CARD  |          |
| Date: Friday, May 26, 2023 | Sheet        | 14 of 21 |

PROPERTY NOTE: this document contains information confidential and property to TOPSTAR and shall not be reproduced or transferred to other documents or disclosed to others or used for any purpose other than that for which it was obtained without the expressed written consent of MPRC

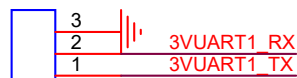
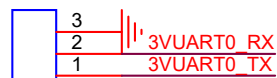
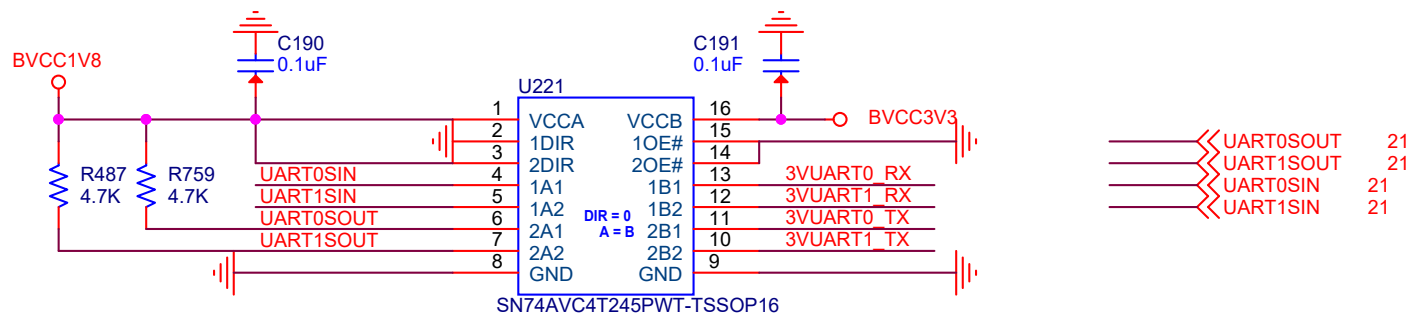


|                                                                                                                                                                                                                                                                                             |              |     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----|
|                                                                                                                                                                                                        |              |     |
| Page Name                                                                                                                                                                                                                                                                                   |              |     |
| IDE Buffer                                                                                                                                                                                                                                                                                  |              |     |
| Size                                                                                                                                                                                                                                                                                        | Project Name | Rev |
| A                                                                                                                                                                                                                                                                                           | V7FPGA CARD  | 1.2 |
| Date: Friday, May 26, 2023                                                                                                                                                                                                                                                                  |              |     |
| Sheet 15 of 21                                                                                                                                                                                                                                                                              |              |     |
| PROPERTY NOTE: this document contains information confidential and property to TOPSTAR and shall not be reproduced or transferred to other documents or disclosed to others or used for any purpose other than that for which it was obtained without the expressed written consent of MPRC |              |     |

## CF卡电路







两个专用串口调试接口  
 注意：它们是3.3V电平



Page Name

UART

Size  
A

Project Name

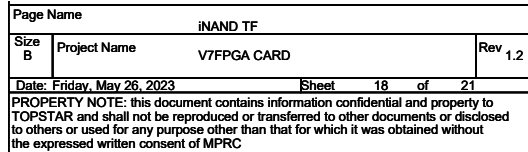
V7FPGA CARD

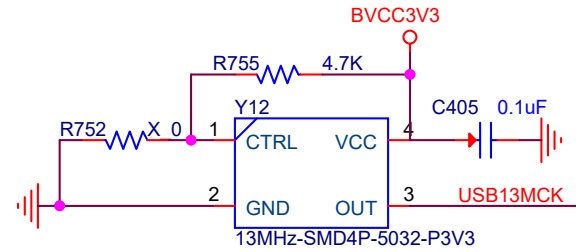
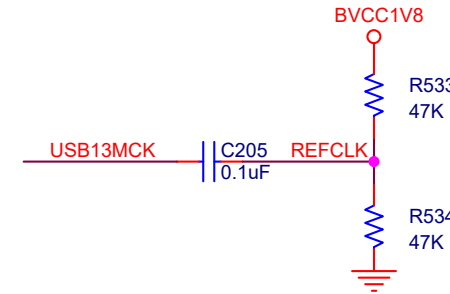
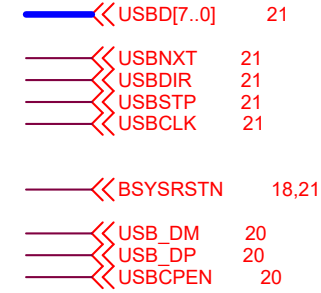
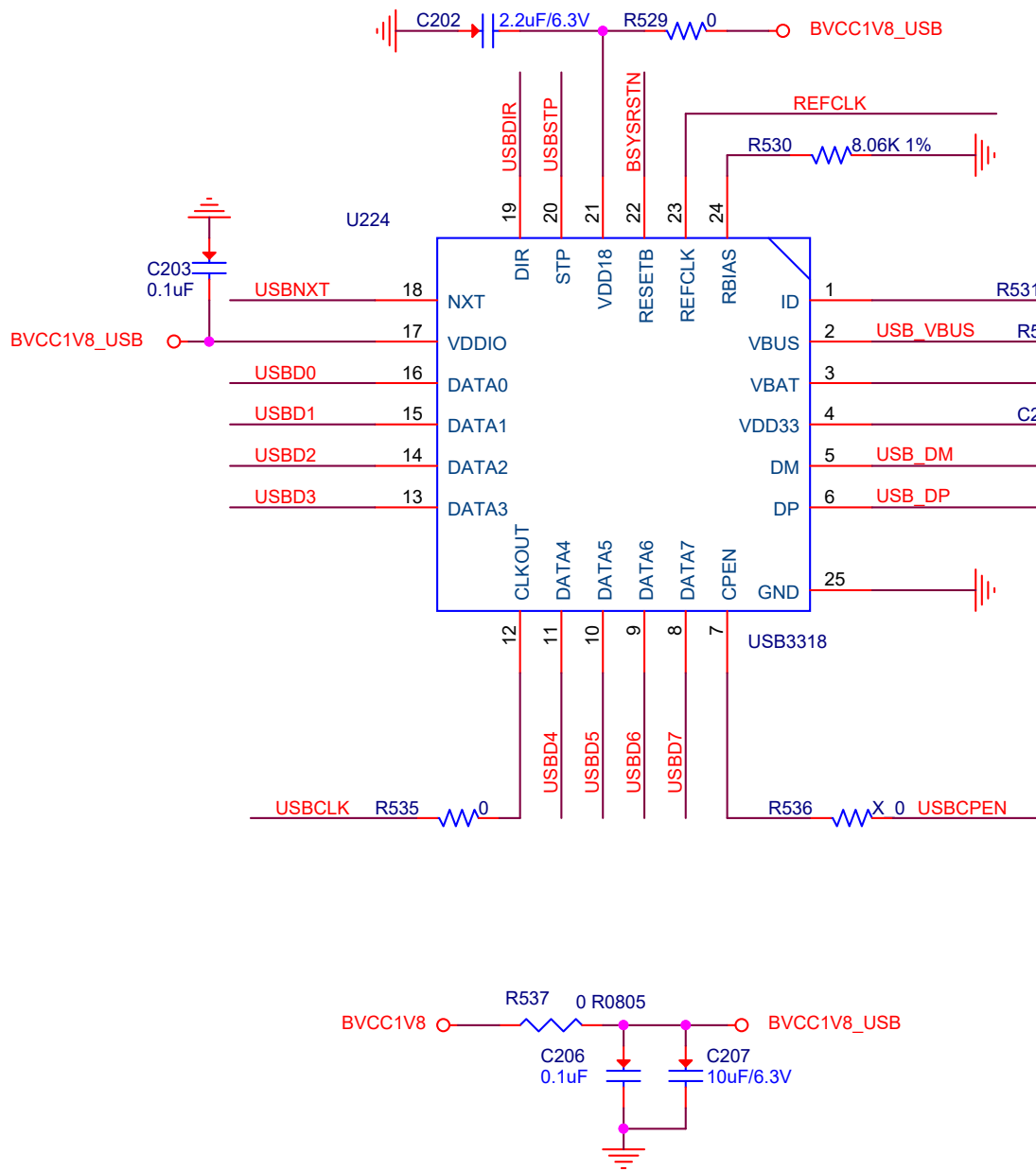
Rev 1.2

Date: Friday, May 26, 2023

Sheet 17 of 21

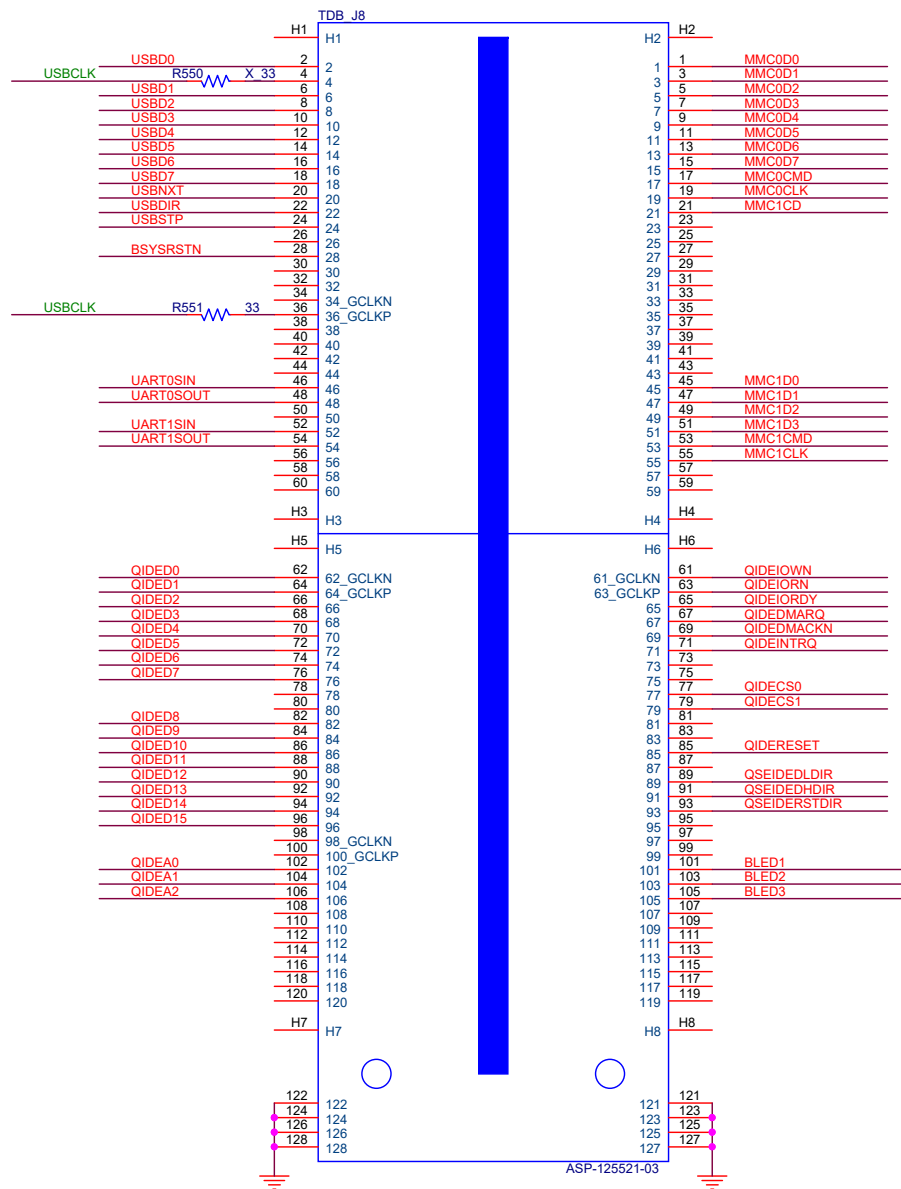
PROPERTY NOTE: this document contains information confidential and property to TOPSTAR and shall not be reproduced or transferred to other documents or disclosed to others or used for any purpose other than that for which it was obtained without the expressed written consent of MPRC



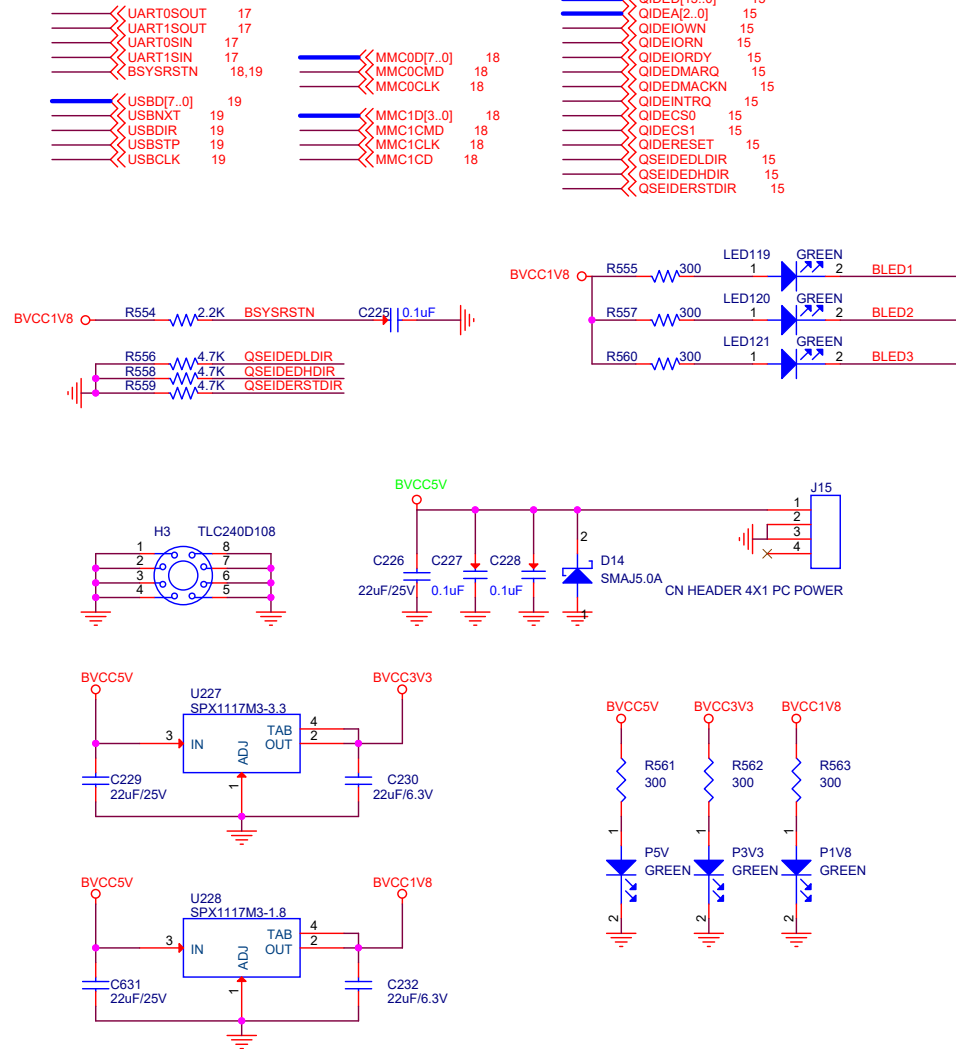
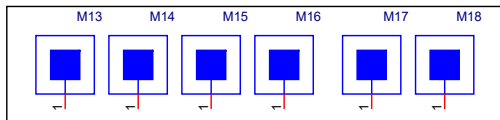


|                                                                                                                                                                                                                                                                                             |              |     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----|
|                                                                                                                                                                                                        |              |     |
| Page Name                                                                                                                                                                                                                                                                                   |              |     |
| USB PHY                                                                                                                                                                                                                                                                                     |              |     |
| Size                                                                                                                                                                                                                                                                                        | Project Name | Rev |
| A                                                                                                                                                                                                                                                                                           | V7FPGA CARD  | 1.2 |
| Date: Friday, May 26, 2023                                                                                                                                                                                                                                                                  |              |     |
| Sheet 19 of 21                                                                                                                                                                                                                                                                              |              |     |
| PROPERTY NOTE: this document contains information confidential and property to TOPSTAR and shall not be reproduced or transferred to other documents or disclosed to others or used for any purpose other than that for which it was obtained without the expressed written consent of MPRC |              |     |





### MARK POINT



|                            |              |                |
|----------------------------|--------------|----------------|
| Page Name                  |              | S2C-DB3        |
| Size                       | Project Name | V7FPGA CARD    |
| B                          |              | Rev 1.2        |
| Date: Friday, May 26, 2023 |              | Sheet 21 of 21 |

PROPERTY NOTE: this document contains information confidential and property to TOPSTAR and shall not be reproduced or transferred to other documents or disclosed to others or used for any purpose other than that for which it was obtained without the expressed written consent of MPRC